



L6 AI Engineer

Module: Responsible AI & Ethics

Topic 4: M6T4:Operationalising Responsible AI

**Welcome to today's
webinar.**

About this topic

This topic aims to:

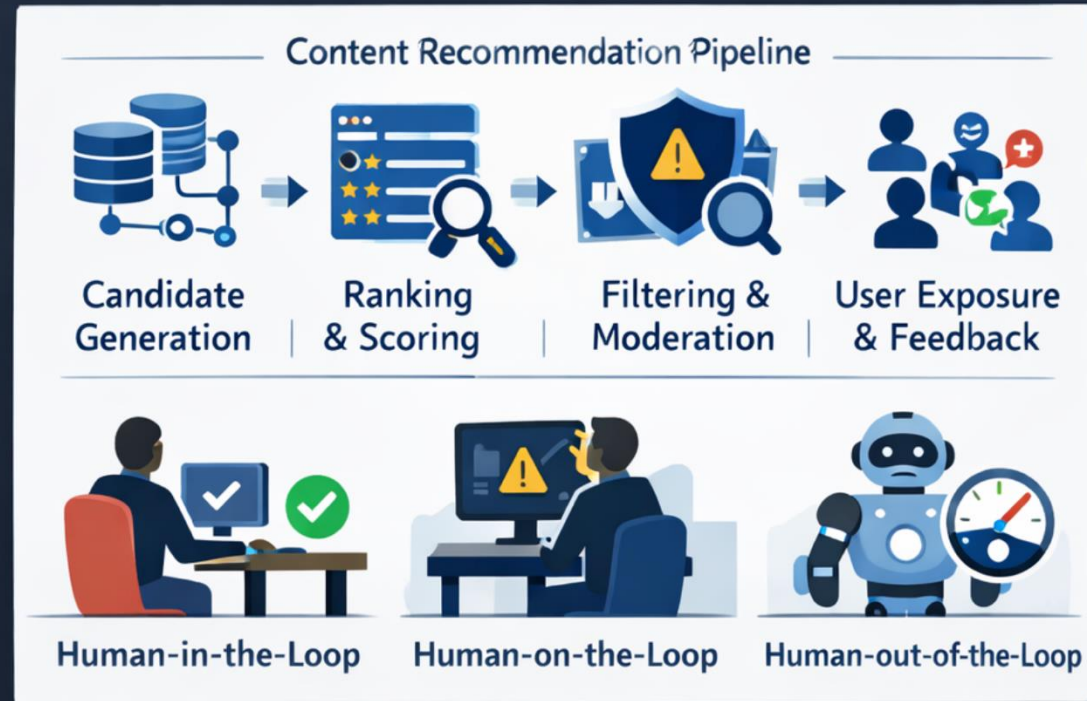
- Explain and apply appropriate human oversight models within operational AI systems.
- Define accountability and incident response processes for AI systems in deployment.
- Apply and evaluate practical mitigation techniques to reduce harmful AI outcomes.
- Analyse AI failures using structured post-incident reasoning and governance controls.
- Integrate ethics, governance, measurement, and mitigation into ongoing Responsible AI operations.

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Human oversight in recommender systems

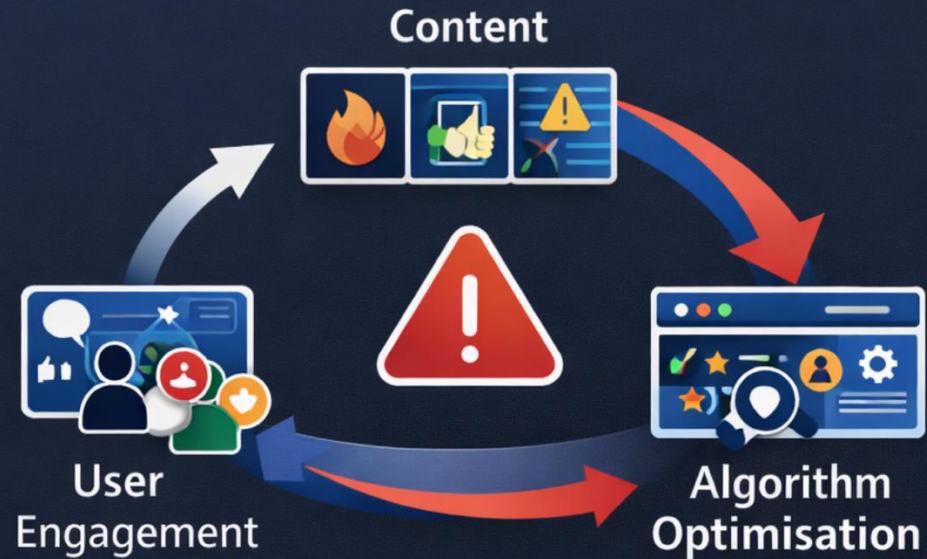
- Why human oversight matters in AI systems at scale
- From Responsible AI principles to operational control
- Focus: content recommendation pipelines



***Oversight** is how organisations remain responsible for outcomes they did not explicitly program.*

Why human oversight is necessary

- Recommender systems operate at scale and speed
- Harm emerges through repetition and amplification
- Optimisation objectives do not equal safety objectives



Systems can cause harm while behaving exactly as designed.

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Human oversight patterns: a spectrum



1 - Human-in-the-loop (HITL)

- Human approves or overrides decisions
- Direct control, slower, higher cost

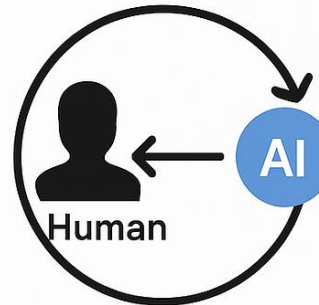
2 - Human-on-the-loop (HOTL)

- System acts autonomously
- Humans monitor, intervene on signals

3 - Human-out-of-the-loop (HOOTL)

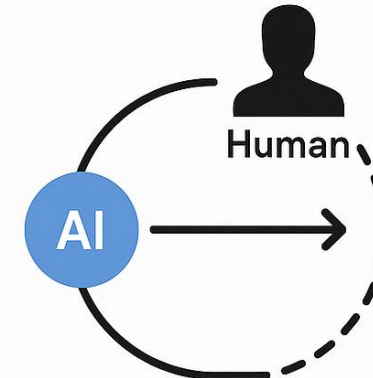
- Fully automated operation
- Oversight occurs after the fact

Human-in-the-Loop

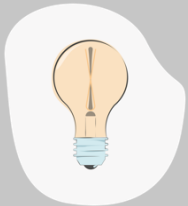
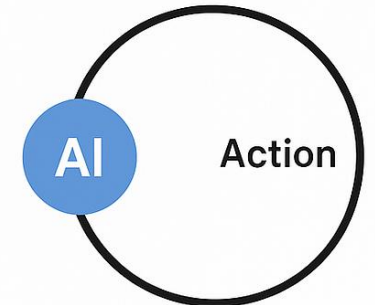


Action

Human-on-the-Loop



Human-out-of-the-Loop



Key Point:

Oversight choice is a design decision, not a moral label.

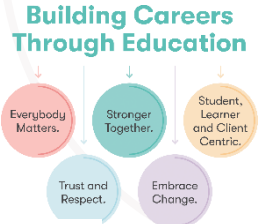
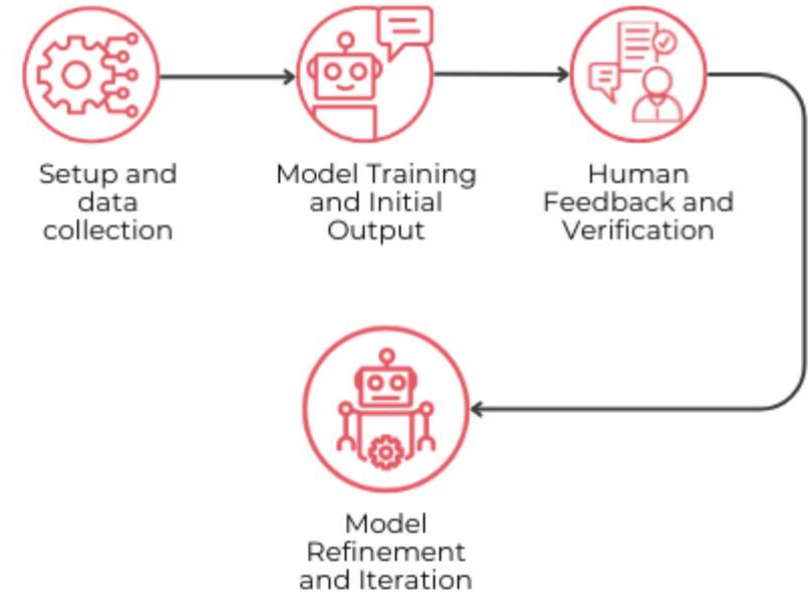
1 - Human-in-the-loop (HITL)

- Human reviews or approves decisions before action
- High control and accountability
- Low scalability and higher cost

Examples in recommender systems:

- Review of flagged or sensitive content
- Appeals and contested recommendations

Human-in-the-loop cycles

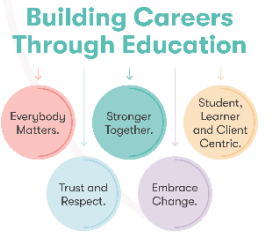
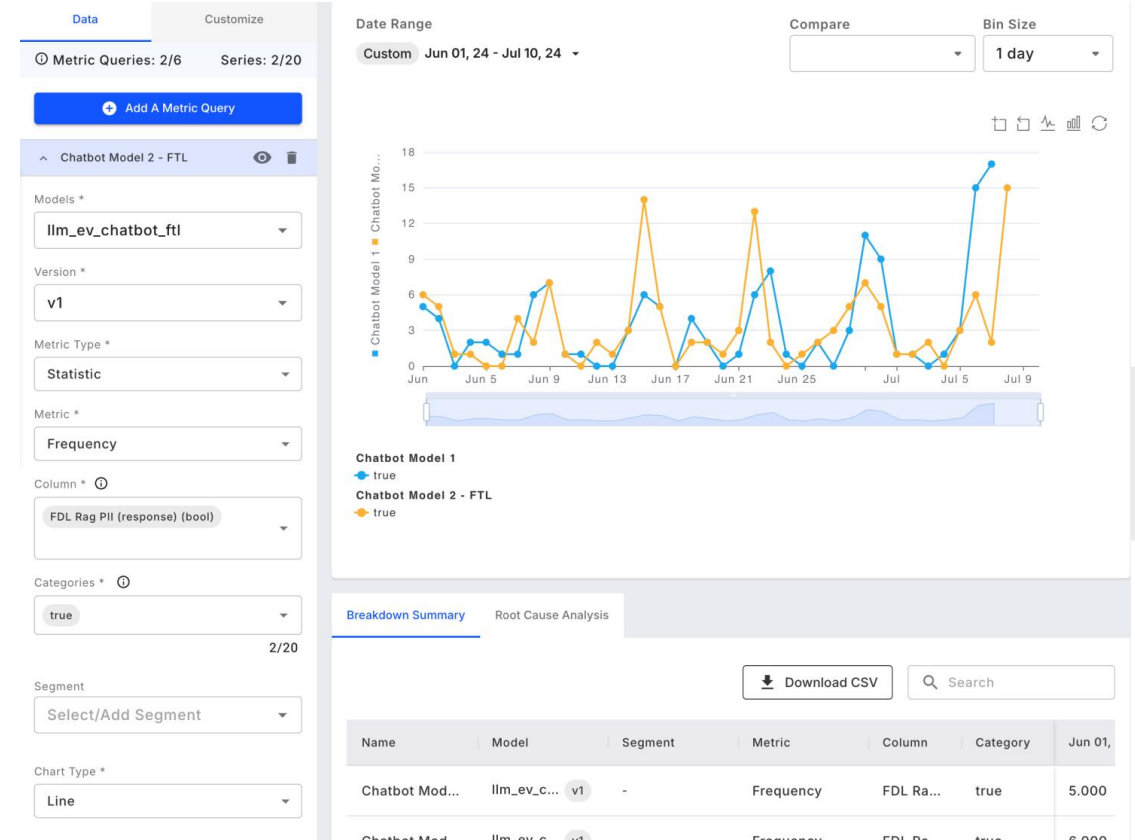


2 - Human-on-the-loop (HOTL)

- System operates autonomously
- Humans monitor behaviour and risk signals
- Intervention triggered by thresholds or alerts

Examples in recommender systems:

- Monitoring harmful-content exposure rates
- Detecting bias, drift, or amplification effects
- Pausing, constraining, or re-ranking recommendations

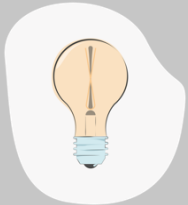


3 - Human-out-of-the-loop (HOOTL)

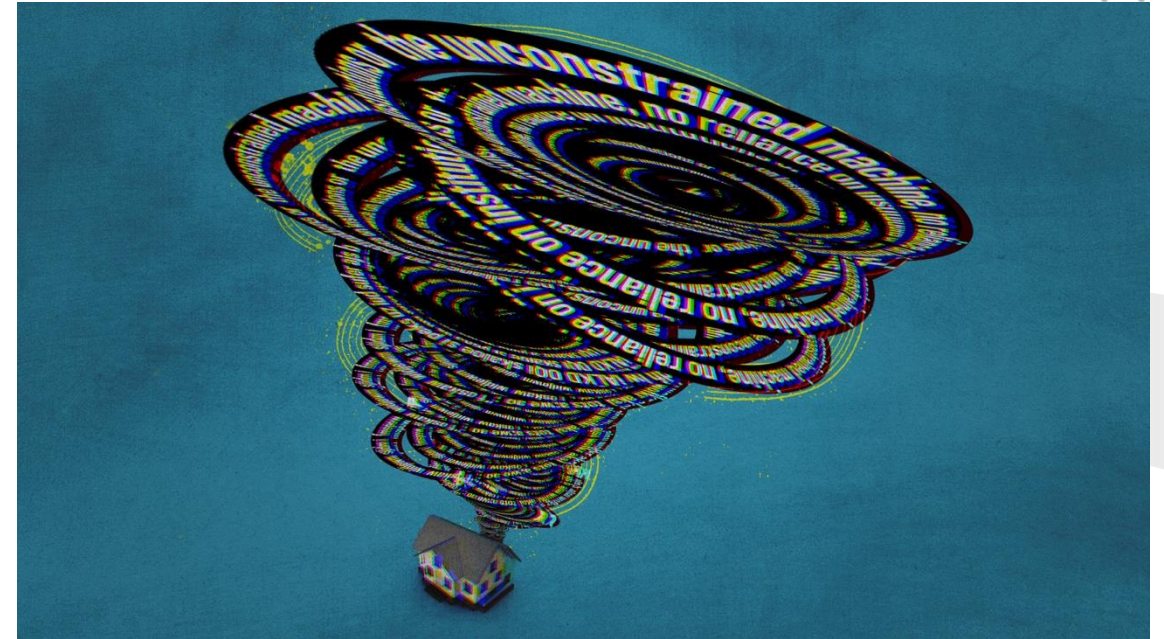
Decisions made entirely by the system
No routine human monitoring or intervention
Maximum speed and scalability

Typical use (limited):

- Low-risk personalisation
- Internal optimisation with reversible impact



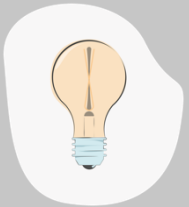
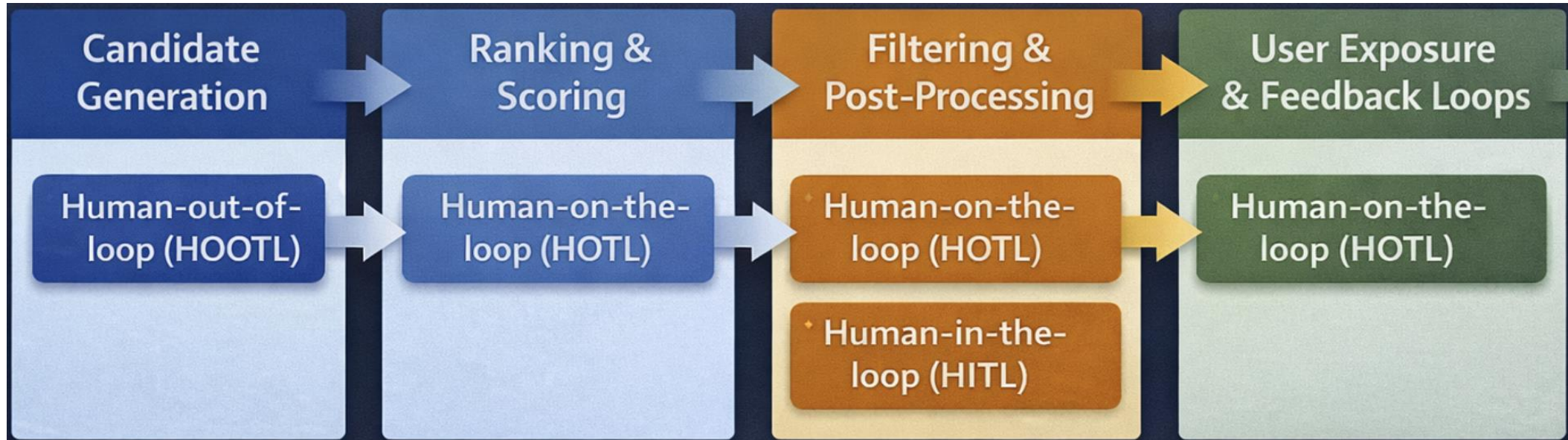
Risk:
Failures can propagate
before anyone notices.



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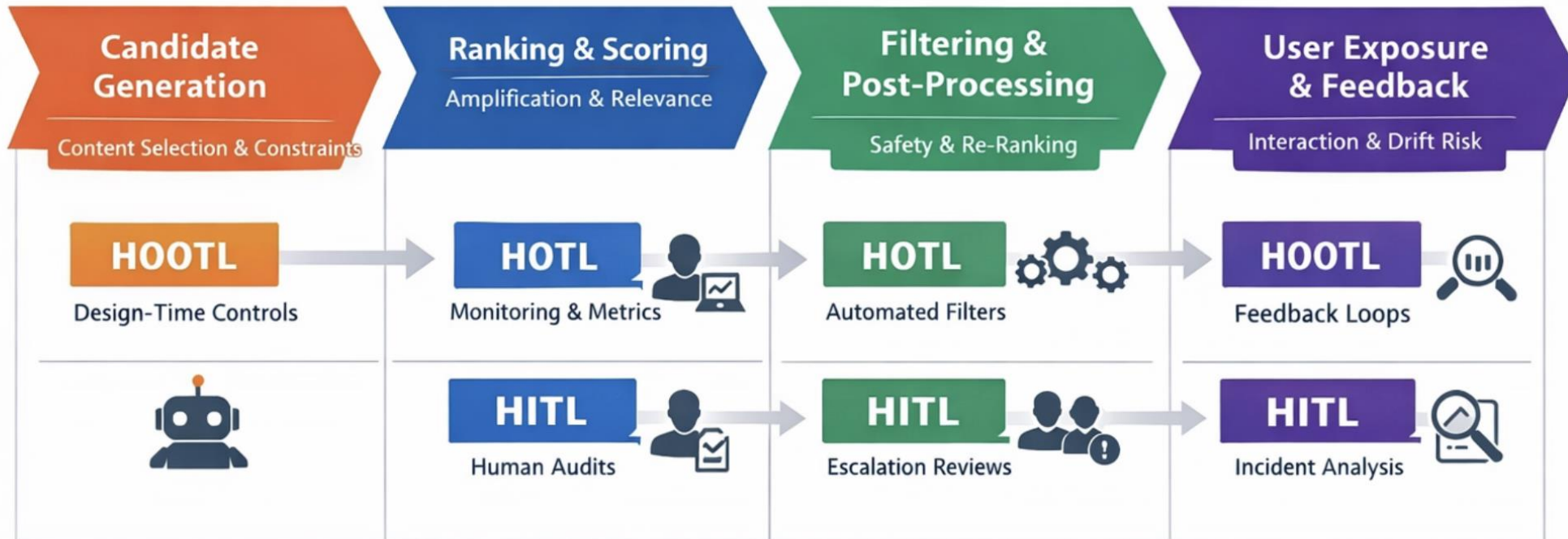


Oversight across the recommendation pipeline



Oversight attaches to stages, not just models.

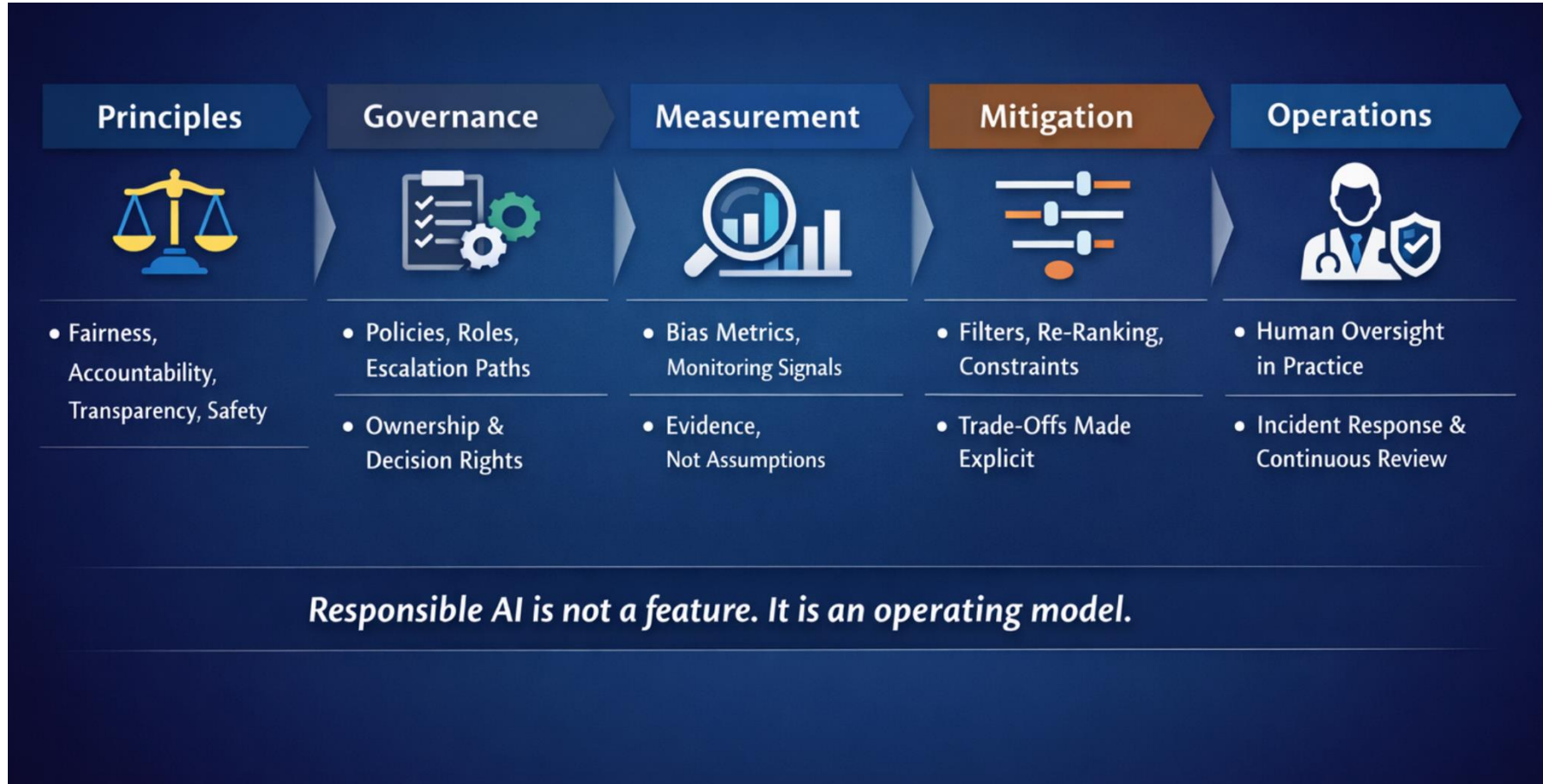
Mapping oversight to pipeline stages



Different risks justify different oversight at different stages.

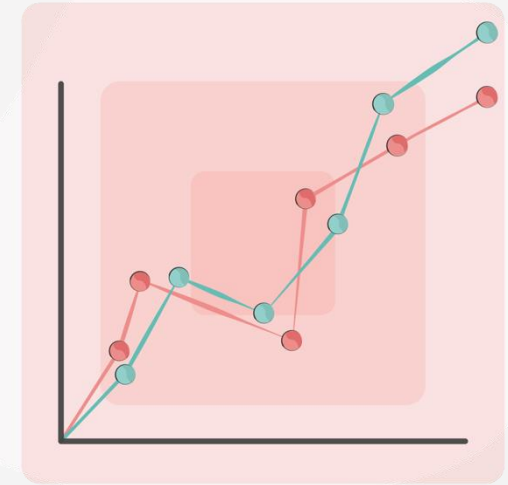
From principles to operations

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Operational Governance



Why governance matters in recommendation systems

- Recommendation systems shape *exposure*, not just decisions
- Harm emerges through scale, repetition, and feedback loops
- Model performance alone does not manage safety risk



Amnesty report finds Facebook amplified hate ahead of Rohingya massacre in Myanmar

World Sep 28, 2022 1:05 PM EDT

With roosters crowing in the background as he speaks from the crowded refugee camp in Bangladesh that's been his home since 2017, Maung Sawyeddollah, 21, describes what happened when violent hate speech and disinformation targeting the Rohingya minority in Myanmar began to spread on Facebook.

"We were good with most of the people there. But some very narrow minded and very nationalist types escalated hate against Rohingyas on Facebook," he said. "And the

By Barbara
Associated

Facebook
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By Amanda
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Facebook's
approved
hate spec
By Barbara
Associated

Go Deeper

Amnesty
Facebook
Meta
Rohingya

Real-World Example: Meta (Facebook / Instagram)

Internal research identified harmful content amplification, but responsibility for exposure-level harm was fragmented across teams.



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When no one owns the outcome, no one can intervene decisively.



Accountability paths: Who owns harm when it occurs?

- Safety risk is organisational, not just technical
- Accountability must be explicit before deployment
- Clear ownership enables timely intervention



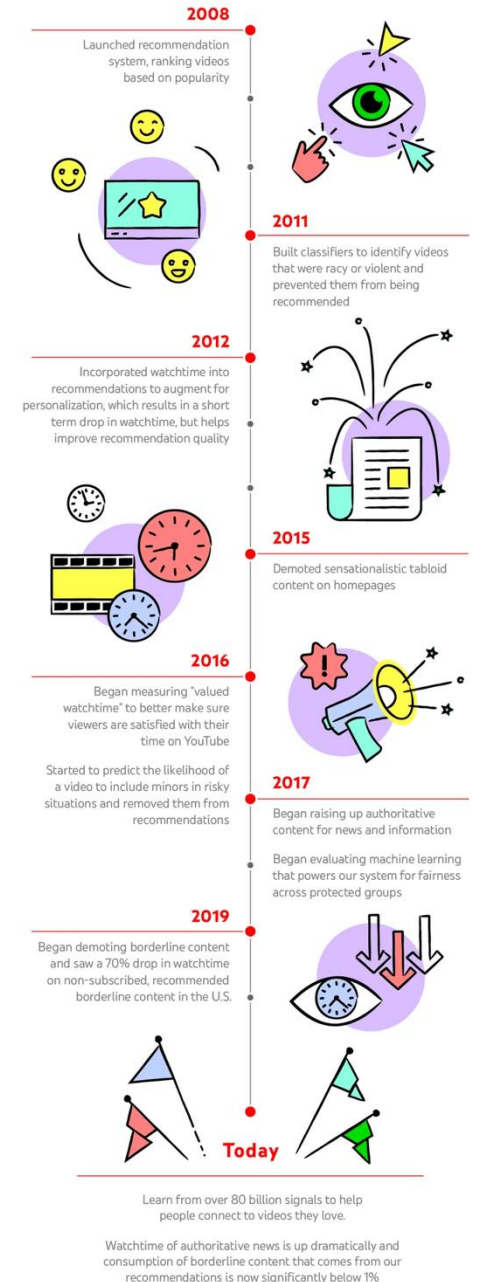
Real-World Example: YouTube

Content moderation and recommendation harms triggered clarification of ownership across **product, policy, trust & safety**, and engineering teams.

If accountability is unclear, response will be slow or contested.

Recommended for You

Key Moments in YouTube's Recommendation System



Incident response for AI systems

- Incidents are socio-technical, not just model failures
- Detection, triage, containment, remediation, review
- Response speed and authority matter more than root-cause perfection

Incident Response Lifecycle



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AI impact assessments: Anticipating harm before deployment

- Structured assessment of potential harms and affected groups
- Goes beyond accuracy, bias, and performance metrics
- Informs go/no-go decisions and safeguards

Real world example: Tiktok

Concerns around adolescent mental health highlighted risks driven by prolonged exposure and feedback loops rather than individual recommendations.



Consider: What happens if this system works exactly as designed?"

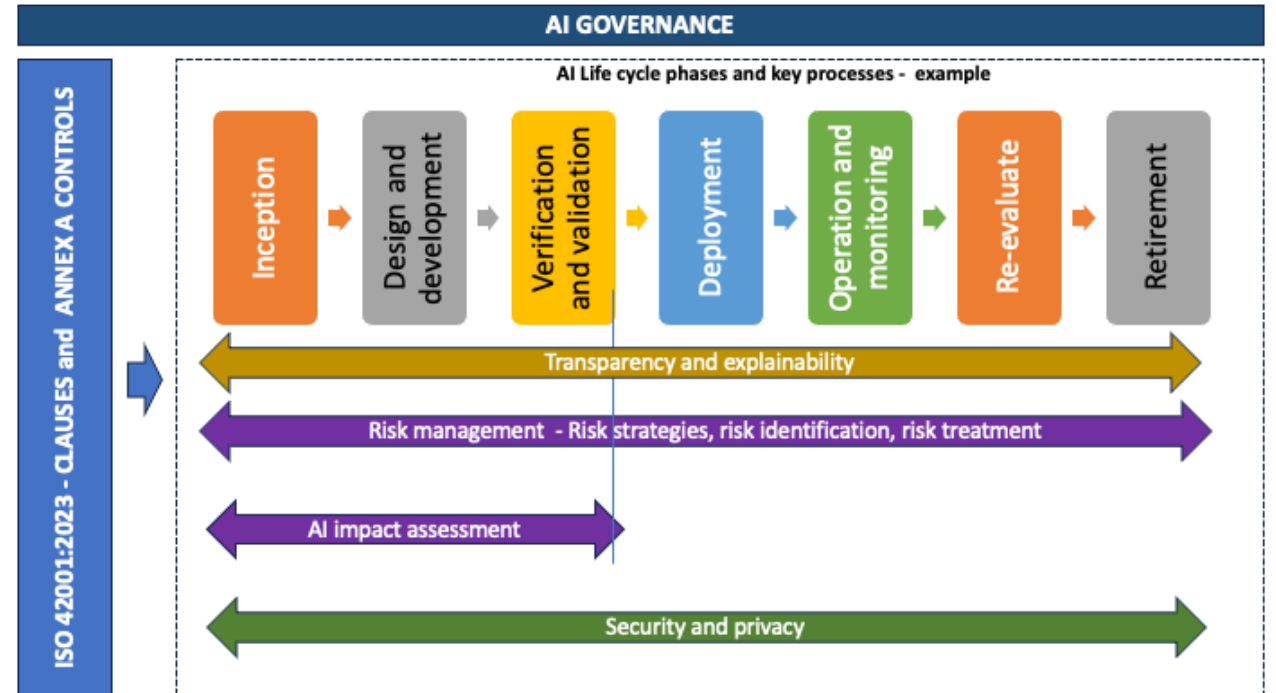


Governing safety risk end-to-end

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- Accountability defines who owns risk and decisions
- Impact assessments anticipate harm before deployment
- Incident response manages harm when it emerges
- Oversight sustains control after launch



From governance to action

- Safety risk in recommender systems is organisational, not purely technical
- Accountability enables intervention
- Impact assessments surface foreseeable harm
- Incident response manages harm under real conditions



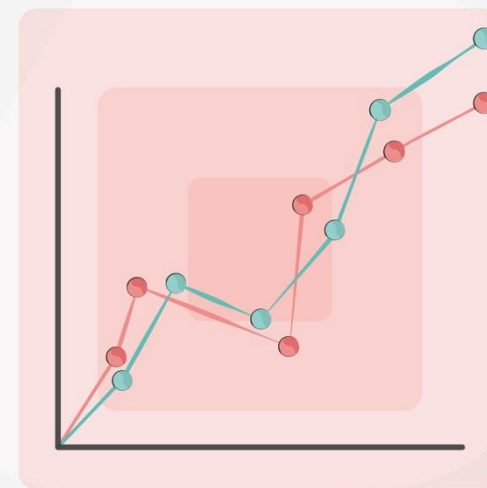
Governance sets the rules - mitigation is where trade-offs become real.

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Lab



Lab brief:

Mitigating harmful content in a recommender system

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What this lab simulates

- A live trust & safety escalation
- A recommender that works technically, but produces unsafe exposure

Your role

- AI assurance / safety team

Your task

- Identify harmful-content exposure
- Apply mitigation without retraining
- Justify relevance–safety trade-offs



Lab Steps

Step 1 – Baseline (≈10 min)

- Measure harmful exposure in top-N rankings

Step 2 – Mitigation (≈15 min)

- Post-processing filter
- Re-ranking (demotion)
- Penalty scoring

Step 3 – Compare (≈10 min)

- Exposure reduction
- Relevance loss
- Rank displacement

Step 4 – Reflect (≈10 min)

- Was the trade-off acceptable?
- Who should approve your thresholds?

Dataset choice: MovieLens 100k

We will be explicit that we are **simulating content risk**, not claiming movies are inherently harmful.

Core tables used

ratings.csv → user–item interactions

movies.csv → metadata (genres)

Synthetic harm labelling We will create a new column:

content_label ∈ {safe, borderline, harmful}

(for example: family files are safe, but horror flagged as harmful)



What we are really testing

This lab is not about

- Building a better model
- Finding a perfect metric

This lab *is* about

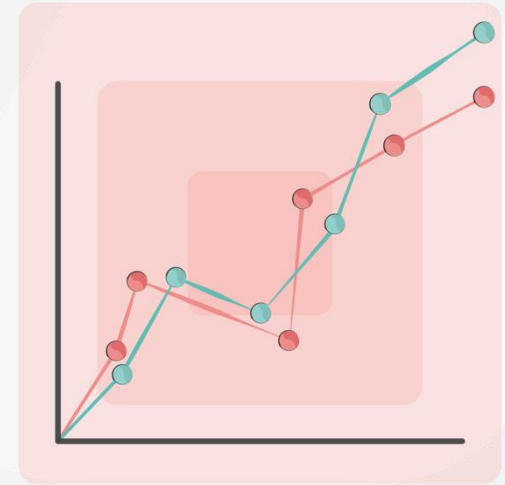
- Governing systems after deployment
- Making value-laden decisions explicit
- Knowing when to escalate to humans

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Group Discussion



Scenario brief: AI system failure



The situation

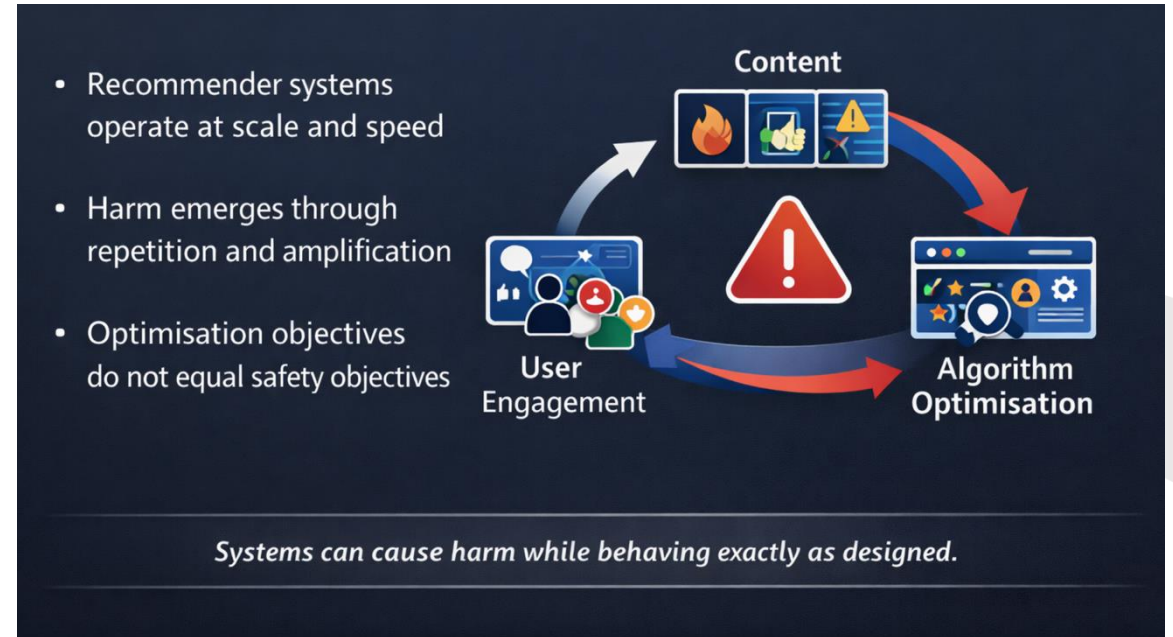
- A recommender system has been live for 6 months
- Engagement metrics are strong
- External scrutiny reveals increased exposure to harmful content

What we know

- No single prediction is “wrong”
- The system behaved as designed
- Internal escalation was slow or unclear

Your role

- Post-incident review panel



Group task: who intervenes, and how?



In your groups, decide:

Detection

- What signals should have triggered concern?
- Where was the first missed escalation point?

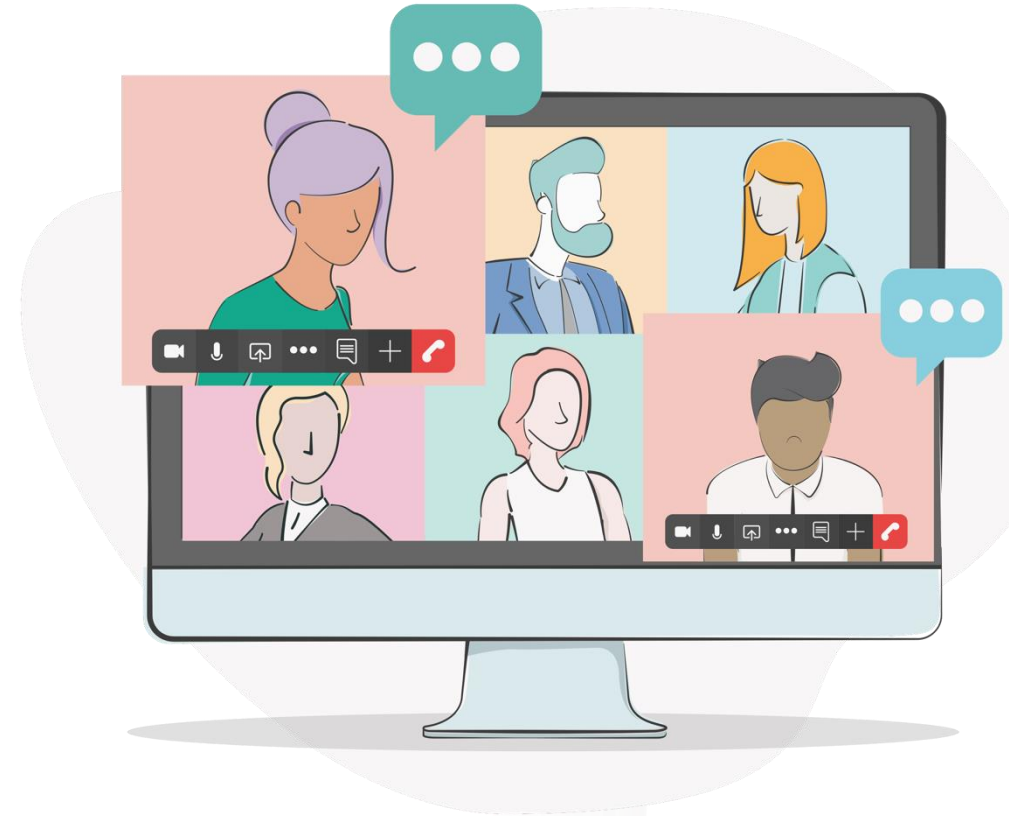
Accountability

- Who owned monitoring?
- Who had authority to intervene?
- Where did accountability break down?

Intervention

- What action should have been taken first?
- What evidence would justify that action?

Use lessons from the lab and earlier segments.



What this exercise is testing

This is not about

- Finding a single mistake
- Designing a perfect control
- Judging individual teams

This *is* about

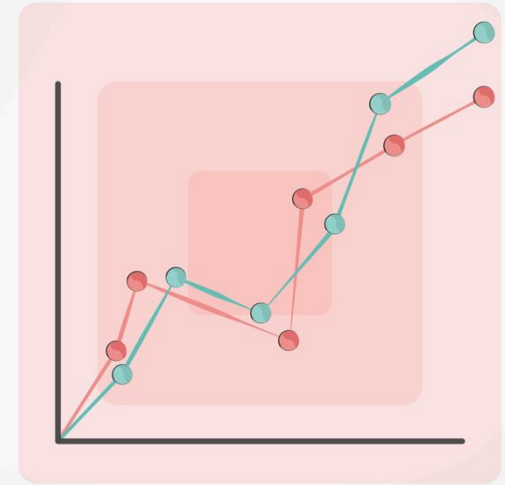
- How responsibility flows in practice
- When technical issues become governance issues
- Why mitigation tools require authority

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Wrap Up

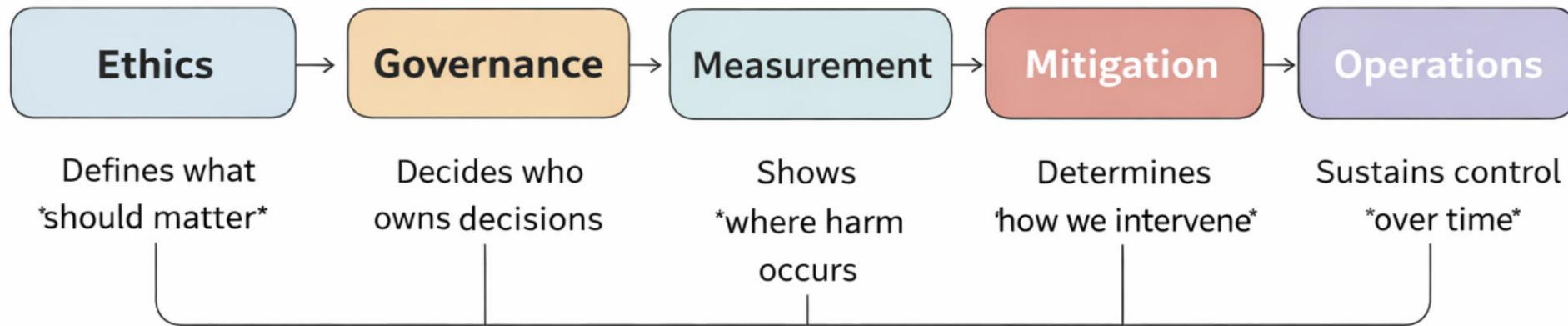


From ethics to operations: the full chain

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Ethics → Governance → Measurement → Mitigation → Operations

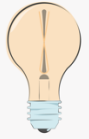


Responsible AI fails when any link in this chain is weak.

Where failures actually occur

Common failure modes

- Ethics without governance → no one acts
- Governance without measurement → blind oversight
- Measurement without mitigation → dashboards, not safety
- Mitigation without authority → delayed or blocked action
- Operations without review → slow, silent drift



Reality check

Most AI failures are not technical failures.



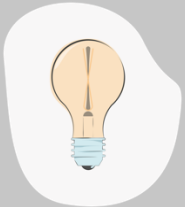
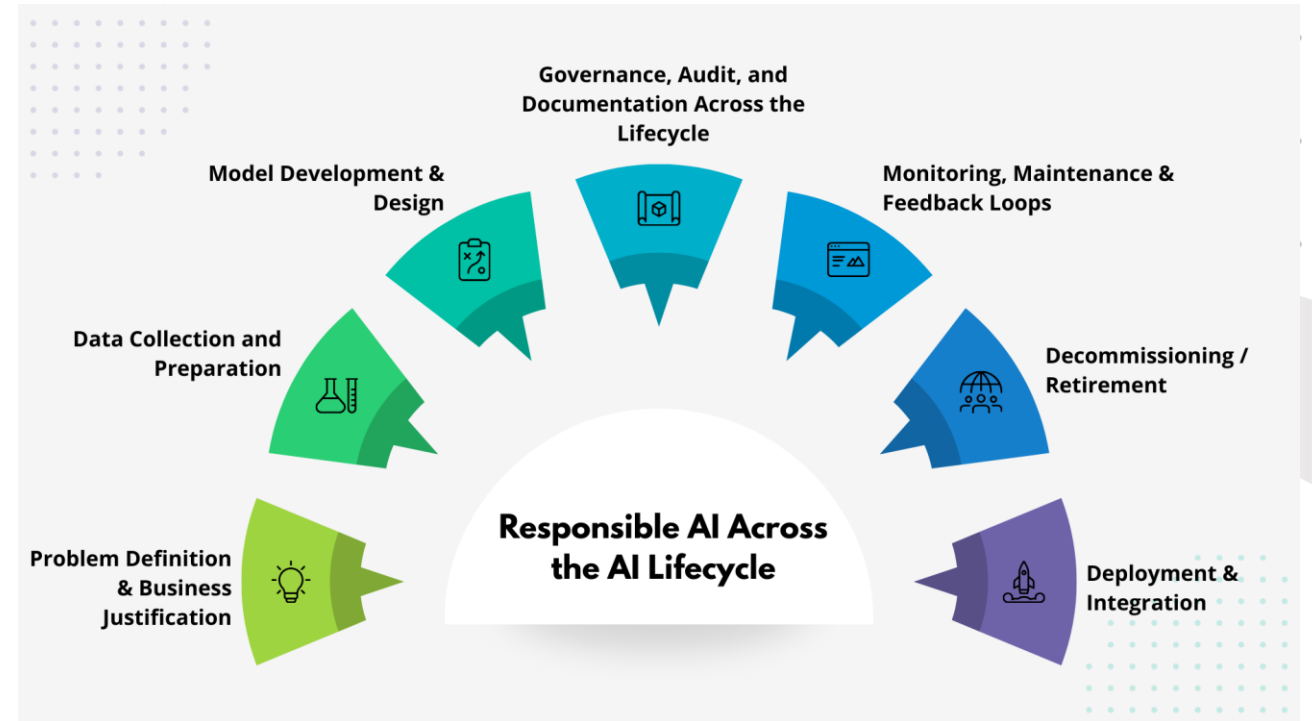
What responsible AI looks like in practice

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Responsible AI in practice means:

- Clear accountability before deployment
- Explicit thresholds for escalation
- Metrics that reflect real harm
- Mitigations that are pre-authorised
- Ongoing review, not one-off approval



Responsible AI is not a property of models.

It is a property of organisations.

Kahoot quiz challenge

Welcome to our end-of-session Kahoot challenge!

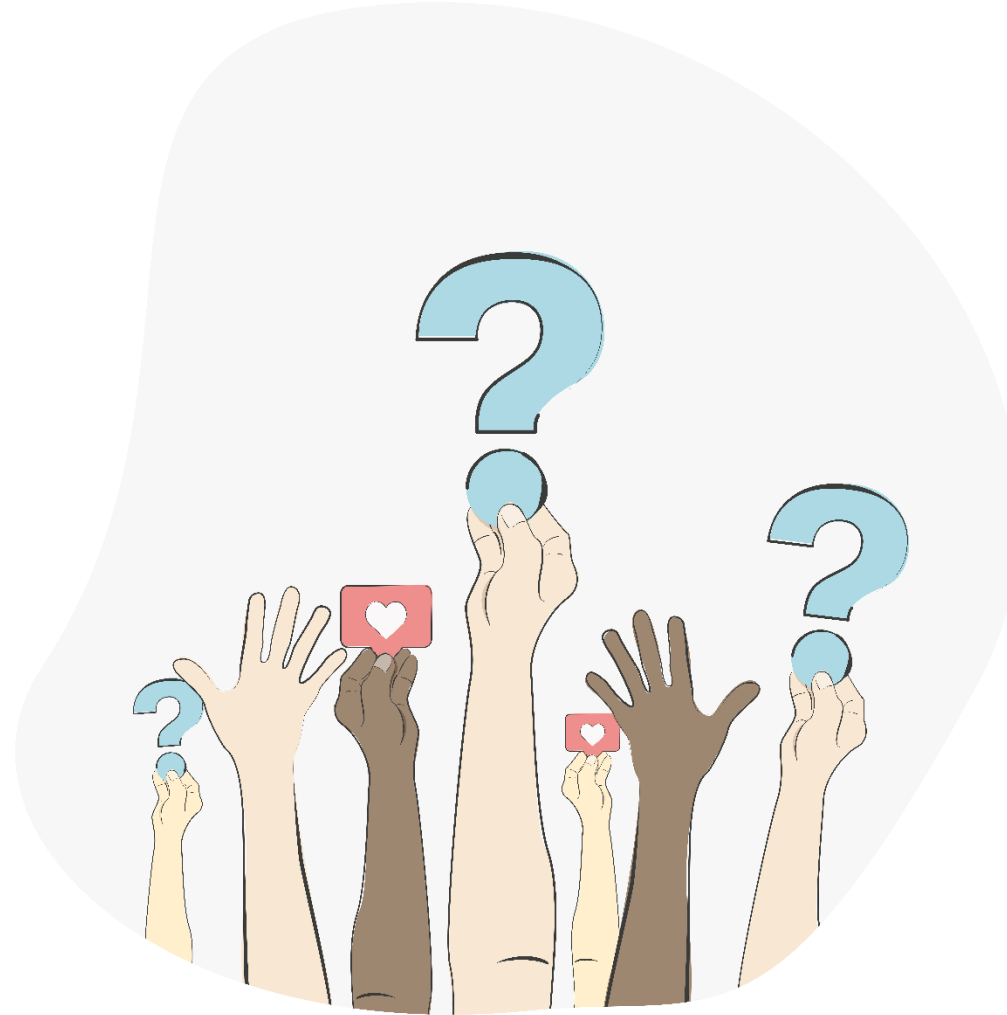
This isn't about catching anyone out.

Think of it as a playful knowledge check to celebrate how much you've absorbed.

Good luck!



Any questions or feedback?



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Thank you

